Related-Work Table: BloodMNIST, Track 3 (Custom CNN + Attention)

CSE475 Machine Learning Project · Task 1 · Dataset: BloodMNIST (8 blood-cell classes)

Scope. Six studies (2022–2025) on blood-cell and biomedical image classification with CNN or attention-based models. The table is split in two parts that share the paper number: Part A describes each paper's data and method, Part B covers its results, limitations, research gap and relation to this project.

Part A: Papers, data and methods

#	Title	Authors	Year	Dataset	Application area	Method / model	Attention type (Track 3)
1	**MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification [1]**	J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, B. Ni	2023	18 standardized biomedical datasets (12 2D, 6 3D), including BloodMNIST: 17,092 images, 8 classes, official 7:1:2 split (11,959 train / 1,712 validation / 3,421 test). Source images: Acevedo et al. [7].	Biomedical image classification benchmark	Standardized lightweight datasets with fixed splits; baselines include ResNet models and AutoML tools (auto-sklearn, AutoKeras, Google AutoML Vision).	None (benchmark paper)
2	**White Blood Cell Classification Using Multi-Attention Data Augmentation and Regularization [2]**	N. Bayat, D. D. Davey, M. Coathup, J.-H. Park	2022	12,444 white blood cell images, 4 classes (lymphocyte, monocyte, eosinophil, neutrophil); random 80/20 split used for the main results.	White blood cell (WBC) subtype classification	CNN backbone (EfficientNet gave the best result) with a multi-attention framework: attention-map generation with a texture-aware residual block, attention regularization, and attention-guided crop/drop augmentation.	Spatial attention maps (custom multi-attention; not CBAM/SE)
3	**An Explainable AI-Based Blood Cell Classification Using Optimized Convolutional Neural Network [3]**	O. Islam, M. Assaduzzaman, M. Z. Hasan	2024	White blood cell (WBC) image dataset.	Blood-cell classification and explainable AI	Optimized CNN with pre-processing (padding, thresholding, erosion, dilation, masking) and hyperparameter tuning; compared with Inception V3, MobileNetV2 and DenseNet201; explained with SHAP, LIME, Grad-CAM and Grad-CAM++; deployed as a web and Android system.	None (Grad-CAM and Grad-CAM++ are visualization tools, not attention modules)
4	**Efficient Leukocytes Detection and Classification in Microscopic Blood Images Using Convolutional Neural Network Coupled with a Dual Attention Network [4]**	S. Khan, M. Sajjad, N. Abbas, J. Escorcia-Gutierrez, M. Gamarra, K. Muhammad	2024	PBC (Acevedo et al. [7]), LISC and Raabin-WBC datasets.	Leukocyte detection and classification (clinical hematology)	CNN coupled with a dual-attention network; DCGAN-generated synthetic images used to address limited training data.	Dual attention network
5	**ConcatNeXt: An Automated Blood Cell Classification with a New Deep Convolutional Neural Network [5]**	M. Erten, P. D. Barua, S. Dogan, T. Tuncer, R.-S. Tan, U. R. Acharya	2025 (online 2024)	17,092 RGB images, 8 classes, from Acevedo et al. [7]; resized to 360×363, network input 224×224; 70:30 train/validation split (authors report about 52.5% train, 22.5% validation, 25% test).	Automated blood-cell classification	Custom CNN (ConcatNeXt) with depth-concatenation blocks, patchify layers and batch normalization; a feature-engineering variant (INCA selection + SVM).	None (Grad-CAM used for visual explanation)
6	**MedNet: A Lightweight Attention-Augmented CNN for Medical Image Classification [6]**	M. Ferdous, S. Mahmud, M. E. Z. Shimul	2025	BloodMNIST, DermaMNIST, OCTMNIST and Fitzpatrick17k.	Multi-domain medical image classification (blood, skin, retina)	Lightweight CNN: depthwise separable convolutions, residual blocks and CBAM attention; Focal Loss; Grad-CAM for interpretation.	CBAM (channel + spatial, hybrid)

Part B: Results, limitations, research gap and relation to this project

#	Paper	Metrics	Strengths	Limitations	Research gap addressed by authors	Relation to your work
1	**MedMNIST v2 [1]**	BloodMNIST, ACC / AUC: ResNet-18 (28) 0.958 / 0.998; ResNet-18 (224) 0.963 / 0.998; ResNet-50 (28) 0.956 / 0.997; ResNet-50 (224) 0.950 / 0.997; auto-sklearn 0.878 / 0.984; AutoKeras 0.961 / 0.998; Google AutoML Vision 0.966 / 0.998. (Number in brackets is input size.)	Standardized, public and reproducible; predefined splits allow fair comparison between methods.	Low default resolution (28×28) may omit fine morphological detail; the BloodMNIST description gives only the 7:1:2 ratio and does not state a patient-level split.	Lack of a broad, accessible, lightweight and standardized benchmark covering diverse 2D and 3D biomedical image-classification tasks.	Source of our dataset and evaluation protocol. We use its official split and its BloodMNIST baselines as reference points.
2	**Bayat et al. [2]**	Accuracy 99.69% (EfficientNet, 80/20 split, 15 epochs); 98.59% with a 60/40 split.	Targets fine-grained differences between similar cell types; combines attention with regularization and attention-guided augmentation.	Covers only the differential WBC count; transferability to other datasets needs further exploration; tested only on CNN backbones.Four-class dataset with a random split, so not comparable with 8-class BloodMNIST.	Distinguishing visually similar WBC subtypes: learning discriminative features and improving training with attention, augmentation and regularization.	Motivates adding an attention module to a custom CNN. We evaluate it separately on the 8-class BloodMNIST task under the official split; its 99.69% is not a comparable score.
3	**Islam et al. [3]**	Test accuracy 99.12%, precision 99%, F1-score 99%.	High accuracy combined with several explanation methods; compared against three transfer-learning models.	Dataset and task differ from BloodMNIST; no attention module, so it gives no evidence about the effect of attention.	The need for accurate WBC classification whose predictions can also be explained, since a high-performing CNN is otherwise hard to interpret.	Supports the explainability part of our project (Grad-CAM, LIME) and the use of precision and F1 alongside accuracy. Serves as a CNN reference rather than a score to compare against.
4	**Khan et al. [4]**	Accuracy: PBC 99.83%, LISC 99.35%, Raabin-WBC 99.60%.	Evaluated on three public datasets; combines attention with synthetic data for the limited-data setting.	Covers detection as well as classification, so the pipeline is more complex than a single custom CNN; splits and protocols differ from the MedMNIST version of the data.	Manual leukocyte counting and inspection is slow and error-prone and annotated data are limited; the authors use attention and synthetic data to improve accuracy, efficiency and generalization.	Attention-based reference on PBC images, the source of BloodMNIST. We test whether a simpler attention-enhanced CNN works on the BloodMNIST version.
5	**ConcatNeXt [5]**	ConcatNeXt: accuracy 97.77%, precision 97.76%, recall 97.83%, F1 97.79%. Feature-engineering variant: accuracy 98.73%, precision 98.85%, recall 98.81%, F1 98.83%.	Custom CNN for the same 8 blood-cell classes; reports precision, recall and F1 as well as accuracy; Grad-CAM explanations.	Uses the original-resolution images (224×224 input) and its own split, not the MedMNIST version, so its numbers are not directly comparable to ours.	Manual blood-cell identification is time-consuming; the authors develop an automated CNN and feature-engineering pipeline.	Closest 8-class CNN reference built on the same source images. Our experiment adds attention to a custom CNN under the official MedMNIST split (different split and resolution).
6	**MedNet [6]**	BloodMNIST: accuracy 0.983 ± 0.0047, AUC 0.998 ± 0.0015 at 384×384 input (mean ± std over three runs).	Lightweight; CBAM vs. non-CBAM ablation across input resolutions; Grad-CAM; tested on several datasets.	*Author-stated:* lower performance on under-sampled classes; error cases include basophils confused with immature granulocytes.384×384 input is costly and not comparable with 28 or 64 px images.	The need for a lightweight CNN that stays accurate and efficient across different medical image datasets by using attention to focus on informative channels and spatial regions.	Closest to our work: a custom lightweight CNN with CBAM on BloodMNIST, with an attention ablation and Grad-CAM. Its 98.3% is a resolution-specific upper reference, not comparable with our 64×64 setting.

References

[1] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni, "MedMNIST v2 – A large-scale lightweight benchmark for 2D and 3D biomedical image classification," *Sci. Data*, vol. 10, art. no. 41, Jan. 2023, doi: 10.1038/s41597-022-01721-8.

[2] N. Bayat, D. D. Davey, M. Coathup, and J.-H. Park, "White blood cell classification using multi-attention data augmentation and regularization," *Big Data Cogn. Comput.*, vol. 6, no. 4, art. no. 122, Oct. 2022.

[3] O. Islam, M. Assaduzzaman, and M. Z. Hasan, "An explainable AI-based blood cell classification using optimized convolutional neural network," *J. Pathol. Inform.*, Dec. 2024.

[4] S. Khan, M. Sajjad, N. Abbas, J. Escorcia-Gutierrez, M. Gamarra, and K. Muhammad, "Efficient leukocytes detection and classification in microscopic blood images using convolutional neural network coupled with a dual attention network," *Comput. Biol. Med.*, vol. 174, art. no. 108146, May 2024, doi: 10.1016/j.compbiomed.2024.108146.

[5] M. Erten, P. D. Barua, S. Dogan, T. Tuncer, R.-S. Tan, and U. R. Acharya, "ConcatNeXt: An automated blood cell classification with a new deep convolutional neural network," *Multimed. Tools Appl.*, vol. 84, no. 20, pp. 22231–22249, 2025 (online Aug. 2024), doi: 10.1007/s11042-024-19899-x.

[6] M. Ferdous, S. Mahmud, and M. E. Z. Shimul, "MedNet: A lightweight attention-augmented CNN for medical image classification," *Sci. Rep.*, vol. 15, art. no. 41936, Nov. 2025, doi: 10.1038/s41598-025-25857-w.

[7] A. Acevedo, A. Merino, S. Alférez, Á. Molina, L. Boldú, and J. Rodellar, "A dataset of microscopic peripheral blood cell images for development of automatic recognition systems," *Data Brief*, vol. 30, art. no. 105474, 2020.

[1]–[6] are the reviewed papers; [7] is the original source of the blood-cell images used by BloodMNIST, PBC and ConcatNeXt.